



Retrieval-Augmented Generation

RAG

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Outline

Where we're going



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Why do we need RAG?

Searching for relevant information in our knowledge base

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RAG pipeline

From documents to embeddings and retrieval

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Parameters

Chunking, overlaps and more...

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RAG in agentic systems

More than just a retrieval

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Limitations of RAG

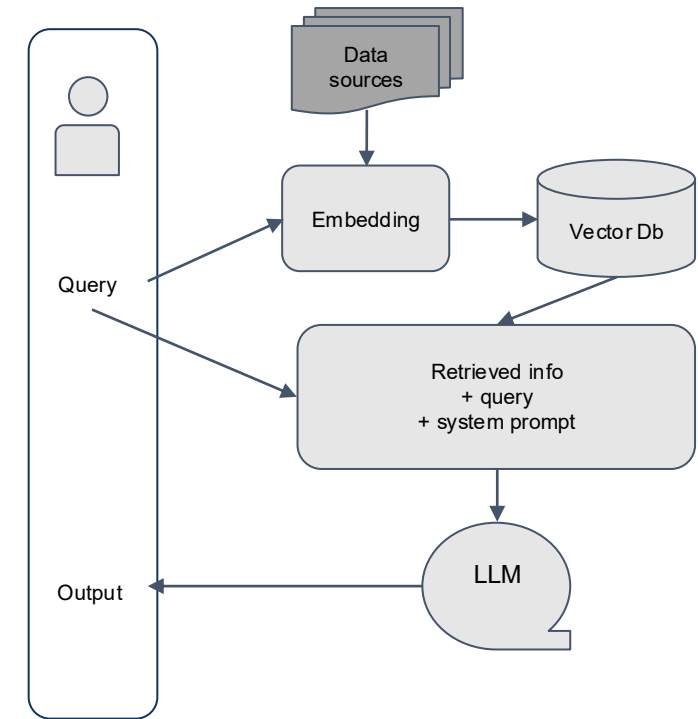
What questions RAG cannot answer

Why do we need RAG

Searching for relevant information in our knowledge base

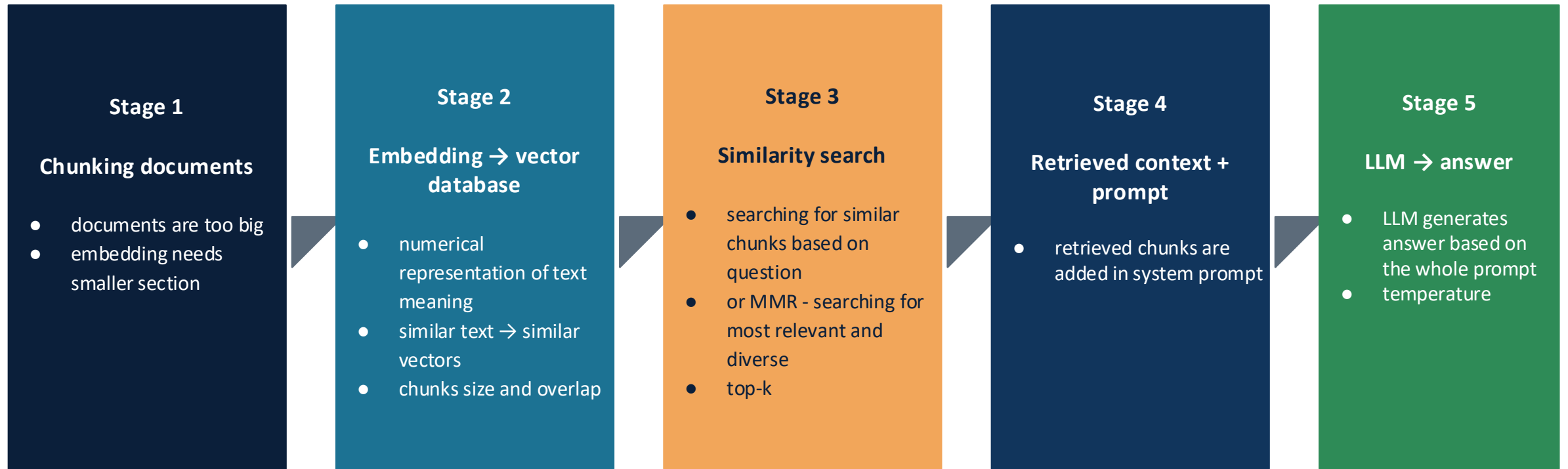
- **chatting over documents**
- **LLM + our knowledge base = RAG**
- **persistent knowledge for the LLM**
- **RAG is not training → LLM gets relevant context**
- **RAG is best for factual question answering over external knowledge**
- **task example:**

**“What is this paper
about?”**



RAG pipeline

From documents to embeddings and retrieval



Parameters

Chunk size

how much text is stored in one retrieval unit

- small chunk size ~ 500
 - precise retrieval
 - less relevant info
 - better for factual lookup
 - loss of context
 - FAQ/support chatbot
- large chunk size ~ 2500
 - more context
 - better coherence
 - more noise
 - retrieval less precise
 - papers/long summaries

Chunk overlap

how much text of the previous chunk repeats inside next chunk

- without overlap: important information can become split across chunks
- with overlap: both chunks contain part of shared context → retrieval becomes more robust
- too little ~ 10 - context fragmentation
- too much ~ 800 - nearly identical chunks

Top-k

how many chunks are retrieved from the vector database

- small top-k ~ 1
 - very focused context
 - lower latency
 - cheaper inference
 - missing info
- large top-k ~ 20
 - broader context
 - better for multi-document reasoning
 - more noise
 - larger prompt cost

Temperature

how random is the LLM output

- this does not affect retrieval
- only affects generation
- for RAG systems is better to set low temperature ~ 0.1
 - deterministic
 - stable
 - less creative
 - → accurate use of retrieved context

→ these parameters are trade offs - there is no universally optimal configuration, it is always based on the type of task you are solving

Limitations of RAG

What questions RAG cannot answer

Context problems

problems in document representation

- ▶ **Information dilution**
 - too much context may overwhelm the model, reduce answer precision and increase hallucinations
- ▶ **Context fragmentation**
 - important information can be separated across the whole document
- ▶ **Chunking problems**
 - too small or too large chunks
 - little overlap - losing the full meaning

Representation quality strongly affects retrieval quality

Retrieval problems

problems during search phase

- ▶ **Semantic mismatch**
 - similarity search retrieves unrelated chunks
 - different wording may produce poor retrieval
 - similarity search works in embedding space, not through true understanding
- ▶ **Retrieval bias**
 - repeatedly retrieving chunks from one document or topic
 - often happens with highly similar documents
- ▶ **Top-k limitations**
 - too few or too many retrieved chunks

RAG system can only reason over what they retrieve

Reasoning problems

problems during inference

- ▶ **Summary problem**
 - RAG is more question and answer tool than summarizing across all documents
- ▶ **Context window limitation**
 - LLMs have finite context windows - model cannot load whole database
 - small subset of knowledge is visible during inference
- ▶ **Lack of planning**
 - no task decomposition
 - no aggregation of evidence
- ▶ **No iterative reasoning**
 - no recognition of missing information
 - no reformulation of failed queries

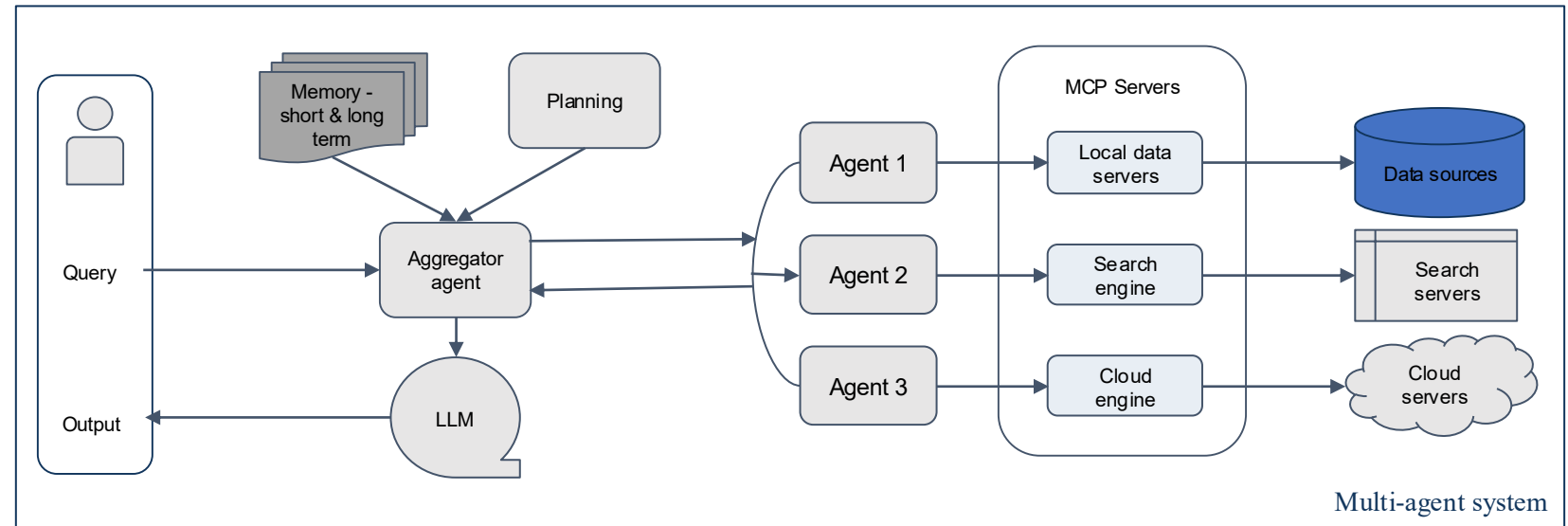
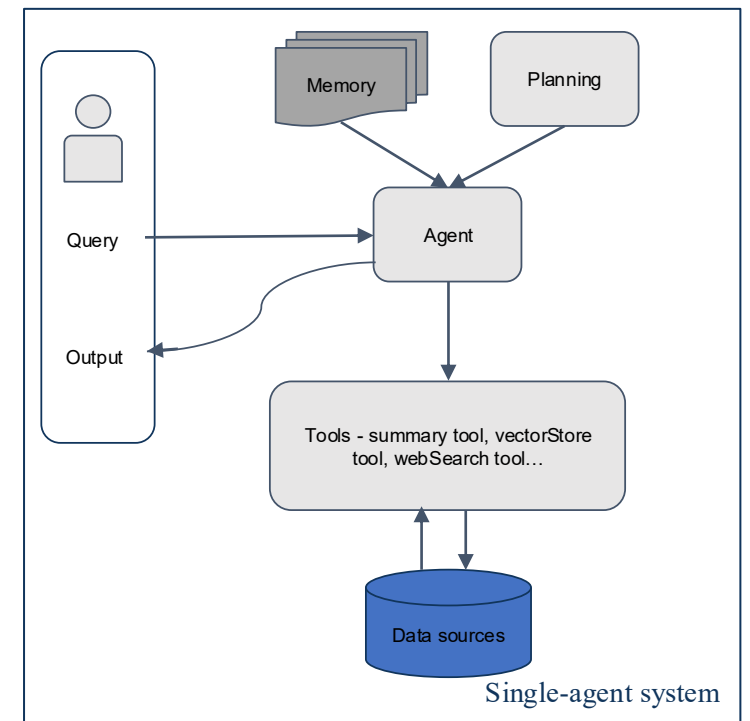
RAG improves access to information but it does not solve all our problems.

RAG in agentic systems

More than just a retrieval

- more retrieval steps - evaluation and refining query
- not just search anymore → better in
 - complex reasoning
 - planning
 - iterative search...
- task example:

“Find contradiction across all uploaded papers”





Summary & Contributions

Understanding the core RAG pipeline

- chunking, embedding, vector database, retrieval (similarity search) and generation
- RAG is not training but it gives LLM extra context

Practical exploration of RAG parameters

- impact of: chunk size, chunk overlap, top-k retrieval and temperature
- parameters directly affect retrieval quality and final answers

Analysis of RAG Limitations

- 3 major categories of RAG limitations
 - retrieval problems
 - context problems
 - reasoning problems
- Basic RAG systems struggle with complex multi-document reasoning tasks

RAG and agentic systems

- modern agentic and multi-agent systems extend traditional RAG
- iterative retrieval, planning, query reformulation, specialized agents

Closing thought. RAG is not about making LLM know everything but giving models access to the right information